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soilphysics: An R package for simulation of soil compaction induced by agricultural field traffic

Renato P. de Lima [a,*], Anderson R. da Silva [b], Álvaro P. da Silva [c]

[a] Department of Agricultural Engineering, Federal Rural University of Pernambuco, Rua Dom Manoel de Medeiros, s/n, Dois Irmãos, CEP: 52171-900, Recife, PE, Brazil
[b] Statistics and Geoprocessing Lab, Agronomy Department, Goiano Federal Institute, Geraldo Silva Nascimento Road, km 2.5, CEP 75790-000, Urutai, GO, Brazil
[c] Department of Soil Sciences, University of Sao Paulo - ESALQ, Av. Padua Dias, 11, CEP 13418-900, Piracicaba, SP, Brazil

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ABSTRACT

Agricultural traffic is the main mechanism responsible for the increase in compacted fields. Strategies to minimize the impact of field traffic on the soil includes knowledge of stress-strain relationships, which involve the use of soil compaction models; however, one of the challenges for dissemination of these models among researchers and academics is its availability as computational tools, easily and freely accessible. In this work, we present a computational tool for simulation of compaction induced by agricultural field traffic based on soil physical and soil mechanics principles, which includes the calculation of contact stress, stress propagation in the soil profile, simulation of bulk density variation in response to the applied stress and assessment of the risk of compaction through readily available soil properties and machinery parameters. Furthermore, the soil inputs of the model is facilitated by a set of pedo-transfer functions for estimation of the required soil properties. The model was computationally implemented as a set of R functions (R software), as well as on an interactive web page, named PredComp, both available in the R package *soilphysics* (available from https://arsilva87.github.io/soilphysics/ or http://cran.r-project.org/web/packages/soilphysics/index.html).

1. Introduction

Soil compaction is a major threat to the main functions of soil and ecosystem services that provide a suitable environment for plant development (Keller et al., 2019). Soil compaction negatively impacts pore systems, soil aeration, water flow and causes an increase in the mechanical impediment to root development (Horn et al., 2019; de Lima et al., 2020a).

Agricultural field traffic is the main mechanism responsible for the increase in compacted areas around the world (Keller et al., 2019; Horn et al., 2019), which is caused by the high stress levels transmitted through tyre-soil interface (Silva et al., 2018). Strategies to minimize the impact of field traffic includes knowledge of stress-strain relationship (O'Sullivan et al., 1999), which on the principles of soil mechanics, comprises the characterization of transmitted stress and compressive behaviour of soil as a function of the stress (O'Sullivan et al., 1999; Horn and Fleige, 2003; Keller et al., 2007).

Tools for characterizing stress-strain relationships include soil compaction models. Some of the recent soil compaction models use different strategies and model structure to evaluate soil compaction: simulation of bulk density variation or assessment of the risk of compaction. The first characterizes the stress transmitted and the variation in soil volume after the applied stress (e.g. *SoilFlex*, Keller et al., 2007). The second characterizes the stress transmitted and compares it with soil strength (usually, the precompression stress), deciding on the risk of compaction (e.g. Terranimo®, Stettler et al., 2014).

Although compaction models are frequently reported in the literature (e.g. Gupta and Larson, 1982; Johnson and Burt, 1990; O'Sullivan et al., 1999; Horn and Fleige, 2003; van den Akker, 2004; Keller et al., 2007, 2015), one of the challenges is still the availability of these models as computational tools that are easily and freely accessible to users. Examples of useful soil compaction models freely accessible are the Terranimo® and *SoilFlex* models, which are available on the webpage and in Excel spreadsheet, respectively. However, the Terranimo® does not yet have the calculation of variation of soil bulk density, whereas *SoilFlex* (Excel spreadsheet) needs to be requested by e-mail.

Sousa et al. (2019) argue that the unavailability of proper open-source computational tools can hinder the propagation of knowledge among researchers and limit the calculation of more complex problems in the soil physical and mechanical fields; and that strategies

* Corresponding author.
E-mail address: renato_agro@alumni.usp.br (R.P. de Lima).

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for dissemination of practical and low-cost tools for soil data analysis should be established. Silva and de Lima (2015); de Lima et al. (2015) and Sousa et al. (2019) suggest that one option is the use of "packages" available freely in the R software (R Core Team, 2020). Silva and de Lima (2015) developed the R package *soilphysics*, which can be freely downloaded in the R software and has a series of useful functions for the soil physics research field (Silva and de Lima, 2015; de Lima et al., 2015; Sousa et al., 2019). Bishwal (2017) describes a set of potential use of R in the field of geoscience, but little has yet been done for the agricultural environments, mainly in relation to soil compaction.

In this work, we present a computational tool for simulation of compaction induced by agricultural field traffic, which includes the calculation of contact stress, stress propagation, simulation of bulk density variation and assessment of the risk of compaction.

2. Theory

2.1. Contact area and stress distribution over the contact area

The formulation described below for the contact area and stress distribution over the contact area follows the semi-analytical model described by Keller (2005). According to Keller (2005), the shape of the contact area can be described by the so-called super-ellipse (Eq. 1):

$$\left|\frac{x}{a}\right|^n + \left|\frac{y}{b}\right|^n = 1 \tag{1}$$

where a and b are the half axes of the super-ellipse and n a positive real number that determines the shape. The area of one quadrant of the supper-ellipse is given by the Eq. 2:

$$CA = b\int_0^a \left(1 - \frac{x^n}{y^n}\right)^{1/n} dx \tag{2}$$

On the tyre-soil interface, considering the super-ellipse, x is the length of the area in the direction of the traffic; y is the tyre width; whereas the n determines the shape of the super-ellipse, as exemplified in Fig. S1. Keller (2005) defined the n parameter empirically as depending on the tyre width (T_{width}) and tyre diameter (T_{diameter}), as given in Eq. 3:

$$n = 2.10(T_{width}T_{diameter})^2 + 2 \tag{3}$$

The width of the contact area (CA_{width}) (where $CA_{\text{width}} = x$; Fig. S1) is given by the T_{width}, whereas the length of the contact area (CA_{length}) (y in Fig. S1) is dependent on the T_{diameter}, tyre inflation pressure (T_{ip}) and recommended inflation pressure (T_{rip}), according to Eq. 4:

$$CA_{length} = 0.47 + 0.11T_{diameter}^2 - 0.16\ln\left(\frac{T_{ip}}{T_{rip}}\right) \tag{4}$$

Since the shape of the contact area is described, the distribution of the stress (σ) over that area, in the y and x directions, can be given from the maximum applied stress, $\sigma_{\max}$. Keller (2005) describes the $\sigma_{\max}$ as a function of T_{ip}, T_{rip} and wheel load (W_{load}) (Eq. 5):

$$\sigma_{\max} = 34.4 + 1.13T_{ip} + 0.72W_{load} - 33.4\ln\left[\frac{T_{ip}}{T_{rip}}\right] \tag{5}$$

The transverse distribution of the σ, i.e. distribution of the σ along the tyre width (x direction) is described by the Eq. 6:

$$\sigma_{(y)} = C\left(\frac{w(x)}{2} - y\right)e^{-\delta\left(\frac{w(x)}{2} - y\right)}; 0 \le y \le \frac{w(x)}{2} \tag{6}$$

where C and δ are parameters and $w(x)$ is the width of contact at position x (e.g. for $x = 0$, $w(x) = T_{\text{width}} = CA_{\text{width}}$). Parameter C is estimated numerically, while δ is given by Keller (2005) empirically as a function of W_{load}, according to Eq. 7:

$$\delta = 9.3 - 7.3W_{load} \tag{7}$$

The value of σ along the driving direction (y direction) is given by the Eq. 8:

$$\sigma_{(x)} = \sigma_x = 0, y\left\{1 - \left(\frac{x}{\frac{l(y)}{2}}\right)^{\alpha}\right\}; 0 \le x \le \frac{l(y)}{2} \tag{8}$$

where $\sigma_x = 0, y$ is the stress under the transverse axis of the tyre footprint, $l(y)$ is the length of contact at position y (for y = 0, $l(y) = CA_{length}$) and α is a parameter (order of the power-law function) that is restricted to $1 \le \alpha \le 16$. The parameter α is calculated under the condition that the modelled wheel load must approximate the weighed (measured) wheel load.

2.2. Stress propagation

An analytical solution for the radial stress propagation (σ_r) under a load point P, acting on a semi-infinite, homogeneous, isotropic and ideal elastic-medium space, was given by Boussinesq (1885) (Eq. 9):

$$\sigma_r = \frac{3P}{2\pi r^2}\cos^3\theta \tag{9}$$

where r is the radial distance from P to the desired point, θ is the angle between the normal load vector and the desired point (Fig. S2).

Based on Boussinesq's equation (1885), Frohlich (1934) introduced (Eq. 10) a concentration factor, ξ, since he noticed that σ measured in the soil did not correspond to the values estimated by Boussinesq's equation (1885):

$$\sigma_r = \frac{\xi P}{2\pi r^2}\cos^{\xi}\theta \tag{10}$$

Söhne (1953) used Eq. 10 to calculate σ_r propagated beneath an agricultural tyre. Söhne (1953) considered that the ξ assumes values of 3, 4, 5, and 6 depending on the soil stiffness, corresponding to very hard, hard, firm and loose, respectively. For ξ = 3, Fröhlich's equation (1934) becomes the equation given by Boussinesq (1885) (Keller, 2004).

Since the distribution of σ over the contact area is not uniform (Söhne, 1953), to calculate the σ_r, Söhne (1953) divided the contact area, A, into small elements, i, of area A_i, and loaded each small area with a load $P_i = A_i\sigma_i$, which is treated as an individual load point (Fig. S2).

The stress at a given depth, z, is calculated by adding the contribution of each P_i. The Söhne's procedure (1953) was used by Keller et al. (2007) in the *SoilFlex* model, and can be defined by Eq. 11:

$$\sigma_r = \sum_{i=0}^{i=n}(\sigma_r)_i = \sum_{i=0}^{i=n}\frac{\xi P_i}{2\pi r_i^2}\cos^{\xi-2}\theta_i \tag{11}$$

Keller et al. (2007) used the procedure defined by Söhne (1953) and calculated the stress components in the x-, y- and z-directions at depth z from Eq. (10) using the Eqs. 12–14:

$$\sigma_z = \sum_{i=0}^{i=n}(\sigma_z)_i = \sum_{i=0}^{i=n}\sigma_{r,i}\cos^2\theta_i \tag{12}$$

$$\sigma_x = \sum_{i=0}^{i=n}\sigma_{r,i}\sin^2\theta_i\cos^2\delta_i \tag{13}$$

$$\sigma_y = \sum_{i=0}^{i=n}\sigma_{r,i}\sin^2\theta_i\sin^2\delta_i \tag{14}$$

where δ is the angle between the shear load vector and the vertical plane.

The main principal stresses σ_1, σ_2 and σ_3 are calculated with help of

the stress invariants I_1 (e.g. Keller et al., 2007) (Eq. 15), whereas the mean normal stress, p, is calculated following O'Sullivan et al. (1999) (Eq. 16):

$$I1 = \sigma_x + \sigma_y + \sigma_z = \sigma_1 + \sigma_2 + \sigma_3 \tag{15}$$

$$p = \frac{1}{3} I_1 = \frac{\sigma_1 + \sigma_2 + \sigma_3}{3} \tag{16}$$

2.3. Stress-strain relationship: bulk density variation

Since the stress at a given soil depth is known, the volume variation can be calculated through the soil compression curve. O'Sullivan and Robertson (1996) proposed a soil compaction model at which the soil volume (V_{soil}) variation is given in the V_{soil}-ln (p) plane, as in the Critical State Theory. For that, the parameters N (the specific volume at $p = 1$ kPa), λ_n (the compression index), κ (the recompression index), κ' (steeper" recompression line) and m (Fig. S3) are required. In Fig. S3, the virgin compression line (VCL), the recompression line (RCL) and the "steeper" recompression line (RCL') are given by the following equations:

$$\upsilon = \left(\frac{\rho_{solid}}{\rho}\right) \tag{17}$$

$$\text{VCL} : \upsilon = N - \lambda_n \ln[p] \tag{18}$$

$$\text{RCL} : \upsilon = \upsilon_i - \kappa \ln[p] \tag{19}$$

$$\text{RCL'} : \upsilon = \upsilon_{yield} - \kappa' \ln[p] \tag{20}$$

The final bulk density (ρ) after the applied stress, p, can be calculated from particle density (ρ_{solid}) and specific volume (υ), using $\rho = \frac{\rho_{solid}}{\upsilon}$ (Keller et al., 2007). The parameter m is dimensionless, and was found experimentally by O'Sullivan and Robertson (1996) to be 1.3. The terms of the Eqs. 17–20 are represented and described in Fig. S3.

2.4. Precompression stress and assessment of the risk of compaction

The tools used to assess the risk of compaction follow a basic principle: if the applied stress is greater than the soil strength, then soil compaction is unavoidable because the soil will unrecoverably deform (Horn and Fleige, 2003; Stettler et al., 2014). Otherwise, if the soil strength is greater than the soil stress, then no permanent deformation will occur and soil damage (i.e. compaction) is not to be expected (Horn and Fleige, 2003; Stettler et al., 2014). Soil strength can be estimated using the precompression stress (σ_p) (e.g. Keller et al., 2011; Schjønning and Lamandé, 2018), which is a measure derived from the soil compression curve and describes the stress point at which the transitions from elastic to plastic behaviour of soils occurs (Stettler et al., 2014; Silva and Lima, 2016; Naderi-Boldaji et al., 2018).

Based on σ_p, Horn and Fleige (2003) and Stettler et al. (2014) proposed methods for assessing the risk of compaction based on ranges of risk, instead of using σ_p as a single value separating the elastic from the plastic behaviour of soil. Horn and Fleige (2003) suggested a classification based on stress (σ) to σ_p relationship, in which for $\sigma/\sigma_p > 1.5$: very stable soil, elastic deformation; $\sigma/\sigma_p = 1.5-1.2$: stable soil; $\sigma/\sigma_p = 1.2-0.8$: labil soil; and $\sigma/\sigma_p < 0.8$: soil weak, additional plastic deformation. The classification of risk suggested by Horn and Fleige (2003) indicates that the greater the σ/σ_p, the lower the risk of compaction.

In the Terranimo® (Terramechanical model) soil compaction model, Stettler et al. (2014) used the σ_p for proposing the following ranges of risk: 1) there is no risk of compaction for $\sigma < 0.5\sigma_p$ (i.e. 50 % σ_p). If $\sigma > 0.5\sigma_p$ and $< 1.1\sigma_p$, there is considerable risk of soil compaction; finally, for σ levels $> 1.1\sigma_p$ (110 % σ_p), considerable plastic deformation (compaction) could be expected (Stettler et al., 2014). These ranges of compaction risk were based on field investigations from Keller et al. (2012), who analysed data from a large number of wheeling experiments and found residual strain lower than the σ_p.

3. Functions architecture

3.1. stressTraffic()

The function stressTraffic() calculates the contact stress and the stress distribution over the contact area, as well as the stress propagation in the soil profile. It needs the inputs required by the semi-analytical model described by Keller (2005) and the concentration factor (ξ) used in Fröhlich's equation (1934). Additionally, the user needs to assign the layers at which the stress propagation will be calculated. For each layer, a corresponding ξ needs to be assigned, which was a procedure used by Horn and Fleige (2003) to distinguish the specific ability of soil layers/horizons concerning the stress compensation. The correspondence among the inputs required by stressTraffic() and the models from Keller (2005) and the Frohlich (1934) are given in Table 1.

The stressTraffic() is also able to provide a graph of the contact area and stress distribution over the contact area through the logical argument plot.contact.area. Furthermore, the output of the stressTraffic() is a list of the contact area (m^2), width (m) and length (m) of the contact area, and the maximum vertical stress, as well as the values of vertical stress and mean normal stress (kPa) at each assigned soil layer.

3.2. soilDeformation()

The function soilDeformation() calculates the variation of soil bulk density after the applied mean normal stress, p, which is calculated using the function stressTraffic(). Additionally, the parameters extracted from the soil compression curve, as described in the model of O'Sullivan and Robertson (1996), need to be assigned for the bulk density variation to be calculated. The parameters (arguments) required by soilDeformation () are described in Table 1. The function also provides a graph for the user to examine the distribution of the VCL, RCL and RCL', such as presented in Fig. S3.

3.3. soilStrength() and compressive_properties()

soilStrength() and compressive_properties() are two groups of functions implemented in *soilphysics* that provide estimates of the σ_p and of the compressive parameters N, λ_n and κ from readily available soil

Table 1
Correspondence among the inputs required by the functions stressTraffic() and soilDeformation() and the models from Keller (2005); Frohlich (1934) and O'Sullivan and Robertson (1996).

Function in the *soilphysics*	Parameter	Inputs in the *soilphysics*	Unit
stressTraffic	T_{ip}	inflation.pressure	kPa
	T_{rip}	recommended.pressure	kPa
	$T_{diameter}$	tyre.diameter	m
	T_{width}	tyre.width	m
	W_{load}	wheel.load	kg
	ξ	conc.factor	–
	Soil layers	layers	m
	p	stress	kPa
soilDeformation	N	N	–
	λ_n	CI	–
	κ	k	–
	κ'	k2	–
	m	m	–
	ρ_{solid}	p.density	Mg m^{-3}
	ρ	iBD	Mg m^{-3}

T_{ip}: tyre inflation pressure; T_{rip}: recommended tyre inflation pressure; $T_{diameter}$: tyre diameter; T_{width}: tyre width; W_{load}: wheel load; ξ: concentration factor; p: mean normal stress; N: the specific volume at $p = 1$ kPa; λ_n: the compression index; κ: the recompression index; κ': the "steeper" recompression line; m: the parameter that separates the virgin compression line and the yield line (Fig. S3).

Table 2
Pedo-transfer functions for estimation of soil precompression stress and compressive parameters from readily available soil properties.

Author	Dependent variable	Explanatory variables	Soil description	Equations
Lebert and Horn (1991)	σ_p (kPa)	ρ: Bulk density (Mg m^{-3}); Lk: Air capacity (v/v, %); nFK: Available water capacity (v/v, %); TW: Non-plant available water capacity (pF > 4.2) (v/v, %); K_s: Saturated hydraulic conductivity ($\times 10^3$) (cm s^{-1}); OM: Organic matter (g 100 g^{-1}); c: Cohesion, kPa; ϕ: Angle of internal friction (°)	Sand	$\sigma_{p1.8} = 438.10\rho - 0.0008\varphi^3 - 3.14\,TW - 0.11(nFK_{1.8})^2 - 465.60$ $\sigma_{p2.5} = 410.75\rho - 0.0007\varphi^3 - 3.41\,TW - 0.35(nFK_{2.5})^2 - 465.60$
			Sandy Loam	$\sigma_{p1.8} = 169.30\rho - 23.03OM^{0.5} + 6.45K_s + 32.18\log(c) - 9.44\varphi + 27.25*\sin(TW) + 119.74\log(nFK_{1.8}) + 19.51$ $\sigma_{p2.5} = 89.50\,\rho - 23.99OM^{0.5} - 2.89K_s + 125.76\log(c) - 1.14\varphi + 26.90*\sin(TW) - 51.46\log(nFK_{2.5}) - 77.25$
			Silt	$\sigma_{p1.8} = 374.15\rho - 4.10OM + 3.38Lk_{1.8} - 1.58K_s^{-0.5} + 1.79c + 1.09\,TW - 6.37\varphi^{0.67} + 0.088(nFK_{1.8})^2 - 472.77$ $\sigma_{p2.5} = 460.71\rho - 20.33OM + 0.088Lk_{2.5} - 2.38K_s^{-0.5} + 2.86c + 4.50\,TW - 20.96\varphi^{0.67} + 0.304(nFK_{2.5})^2 - 610.62$
			Clay<35	$\log(\sigma_{p1.8}) = 0.843\rho - 0.544K_s^{0.33} - 0.022\,TW + 7.03c^{-1} + 0.024\varphi - 0.015\,nF\,K_{1.8} + 0.725$ $\log(\sigma_{p2.5}) = 0.844\rho - 0.456K_s^{0.33} - 0.026\,TW + 12.88c^{-1} + 0.003\varphi - 0.016\,nF\,K_{2.5} + 1.419$
			Clay>35	$\sigma_{p1.8} = 4.59\rho - 1.02OM - 16.43K_s^{0.33} + 0.31\,TW - 1.57\,nF\,K_{1.8} + 3.55c + 1.18\varphi - 18.03$ $\sigma_{p2.5} = 70.65\rho - 0.55OM - 7.01K_s^{0.33} + 1.32\,TW - 1.08\,nF\,K_{2.5} + 3.55c + 1.05\varphi - 100.94$
Imhoff et al. (2004)	σ_p (kPa)	ρ: Bulk density (Mg m^{-3}); CC: clay content (g 100 g^{-1}); w: water content (g g^{-1})	Hapludox	$\sigma_p = -566.764 + 442.891\rho + 4.338CC - 773.057w$
Saffih-Hdadi et al. (2009)	σ_p (kPa)	ρ: Bulk density (Mg m^{-3}); w: water content (g g^{-1}, %)	Very fine	$\sigma_p = 7.71 + 112.21\rho - 2.81w$
			Fine	$\sigma_p = 4.19 + 202.54\rho - 10.92w$
			Medium fine	$\sigma_p = -223.71 + 347\rho - 7.93w$
			Medium	$\sigma_p = -136.87 + 155.19\rho$
			Coarse	$\sigma_p = -220.68 + 191.45\rho + 2.77w$
Severiano et al. (2013)	σ_p (kPa)	h: Matric suction (kPa)	Oxisols for a range of clay content from 152 to 521 g kg^{-1}	$\sigma_p(_{152}) = 129\,h^{0.15}$ $\sigma_{p(209-263)} = 123\,h^{0.13}$ $\sigma_{p(311)} = 85\,h^{0.17}$ $\sigma_{p(365-490)} = 70\,h^{0.16}$ $\sigma_{p(521)} = 63\,h^{0.15}$
Schjønning and Lamandé (2018)	σ_p (kPa)	ρ: Bulk density (Mg m^{-3}); CC: Clay content (g g^{-1}); pF = log10 (h) (h: Matric suction, hPa)	Luvisol	$\sqrt{\sigma_p} = -26.51 + 25.57\rho - 20.40CC + 10.26CC \times pF$
O'Sullivan et al. (1999)	N, λ_n, κ	w: water content (g g^{-1}, %)	Sandy Loam	$N = 2.430 - 0.0055(w-11.2)^2$ $\lambda_n = (N-1.572)/17$ $\kappa = \lambda_n(0.119 - (0.082w/17))$
			Clay Loam	$N = 2.813-0.0128(w-17.4)^2$ $\lambda_n = (N - 1.557)/26$ $\kappa = \lambda_n(0.119 - (0.082w/26))$
Defossez et al. (2003)	N, λ_n	w: water content (g g^{-1}, %)	Loess	$N = 1.9997 + 0.2629\,w - 0.02753\,w^2 + 0.00111\,w^3 - (1.6 \times 10^{-5})\,w^4$ $\lambda_n = 0.1402 + 0.00192\,w + 0.00021\,w^2 - (1.3 \times 10^{-5})\,w^3$
			Calcareous	$N = 2.4196 + 0.13767\,w - 0.02035\,w^2 + 0.00121\,w^3 - (2.4 \times 10^{-5})\,w^4$ $\lambda_n = 0.08051 + 0.03131\,w - 0.00502\,w^2 + 0.00031\,w^3 - (6 \times 10^{-6})\,w^4$
Keller and Arvidsson (2007)	N, λ_n	v_0: specific volume	All soils	$N = -0.007 + 1.196v_0$ $\lambda_n = -0.182 + 0.212v_0$
de Lima et al. (2018)	N, λ_n, κ	ρ: Bulk density (Mg m^{-3}); pF = log10(h) (h: Matric suction, hPa)	Sandy Loam	$N = 4.30 - 1.697\rho + 0.307\,pF - 0.064pF^2$ $\lambda_n = 0.2742 - 0.174\rho + 0.067\,pF - 0.014pF^2$ $\kappa = 0.056 - 0.044\rho + 0.019\,pF - 0.003pF^2$
			Sandy Clay Loam	$N = 4.036 - 1.727\rho + 0.521\,pF - 0.100pF^2$ $\lambda_n = 0.137 - 0.158\rho + 0.150\,pF - 0.029pF^2$ $\kappa = 0.051 - 0.052\rho + 0.031\,pF - 0.006pF^2$
de Lima et al. (2020b)	N, λ_n, κ	pF = log10(h) (h: Matric suction, hPa)	Sandy Loam (Plough layer)	$N = 1.4612 + 0.5080\,pF - 0.0723pF^2$ $\lambda_n = 0.0233 + 0.0728\,pF - 0.0110pF^2$ $\kappa = 0.0221e^{pF(-0.4530)}$
			Sandy Loam (Plough pan)	$N = 1.2726 + 0.4223\,pF - 0.0691pF^2$ $\lambda_n = 0.0112 + 0.0578\,pF - 0.0105pF^2$ $\kappa = 0.0252e^{pF(-0.5414)}$

properties. With soilStrength(), it is possible to estimate, for example, σ_p using pedo-transfer functions from Lebert and Horn (1991); Imhoff et al. (2004); Saffih-Hdadi et al. (2009); Severiano et al. (2013) and Schjønning and Lamandé (2018). Furthermore, using the function compressive_properties(), soil compressive parameters can be estimated by pedo-transfer functions developed by O'Sullivan et al. (1999); Defossez et al. (2003); Keller and Arvidsson (2007); de Lima et al. (2018) and de Lima et al. (2020b). In *soilphysics*, each pedo-transfer function for estimation of the σ_p and compressive parameters is numerically identified, i.e. soilStrength(), soilStrength2() and described for their respective authors. Required inputs are described on the help page. Table 2 shows the list of pedo-transfer functions used to estimate σ_p and compressive properties.

4. Illustration examples

With *soilphysics*, the user is able to follow two paths to evaluate soil compaction: variation of soil bulk density or an assessment of the risk of compaction, which are strategies used by the soil compaction model *SoilFlex* (Keller et al., 2007) and Terranimo® (Stettler et al., 2014), respectively. The first alternative includes using the functions stressTraffic() and soilDeformation() (for which the inputs can be estimated using the function compressive_properties()), whereas for the second strategy, the user must use the functions stressTraffic(), followed by the function soilStrength(), or independently, assign values of σ_p, instead using soilStrength(). If the user would like to estimate only the applied stress over the contact area and the stress propagation, then only the function stressTraffic() shall be used.

To illustrate stressTraffic() and soilDeformation(), we used the machinery parameters available in de Lima et al. (2020a), who performed a wheeling experiment under no-till system computing the effect of the passage of the front and rear tyres of a maize harvester. In de Lima et al. (2020a), measurements and simulated values of soil bulk density after wheeling at a depth of 0.05 m were compared using the same set of functions and modeling as this example. The required mechanical input parameters N, λ_n, κ were obtained by applying the pedo-transfer function (sandy loam soil) developed by de Lima et al. (2018) (Table 2) for an initial bulk density and pF of 1.50 Mg m^{-3} and 2.08 (de Lima et al., 2020a) (Table 3), respectively. After the passage of the front tyre, the mechanical input parameters were recalculated (using the pedo-transfer function) considering the outputted final bulk density (see Table 3), which was used as input for the simulation of the passage of the rear tyre. The parameters m and κ' were assigned following O'Sullivan and Robertson (1996): $m = 1.3$ and $\kappa' = (\lambda_n \kappa)^{0.5}$ for both tyres. To exemplify, we extended the stress propagation to 1.0 m in depth, and to follow consistent with the calculations from de Lima et al. (2020a), we used $\xi = 5$. However, we simulated a slight harder layer ($\xi = 4$) at a depth from 0.30 to 0.40 m to show the effect of ξ (e.g. Horn and Fleige, 2003). To simplify, the same initial compressive parameters were assigned to all simulated soil layers.

In the second set of simulations, we assessed the risk of compaction using stress-strength relationships. To associate the results with the first set of simulations, the risk of compaction was analyzed in terms of mean normal stress, p. For that, we extracted σ_p from the soil compression curve for the initial soil condition from the wheeling experiment from de Lima et al. (2020a) (i.e. bulk density of 1.50 Mg m^{-3} and pF = 2.08; Table 3) through the intersection of the RCL with the VCL (Fig. S3), which was equal to 81 kPa. Precompression stress was compared to the stress induced by the front tyre of the maize harvester, which was the tyre estimated to apply the greater mean normal stress (de Lima et al., 2020a).

5. Results

Fig. 1 shows the contact area and stress distribution over the contact area, as well as the vertical and mean normal stresses propagation calculated using stressTraffic() for the front and rear tyres configurations given in Table 3. Larger maximum contact stress (front tyre = 278 kPa; rear tyre = 209 kPa), contact area (front tyre = 0.52 m^2; rear tyre = 0.17 m^2) and stress propagation were observed for the front tyre in response to tyre configuration (see Keller, 2005; Schjønning et al., 2008). At a depth of 0.3–0.4 m, there is a stress deviation caused by the changes in the concentration factor (ξ) (Fig. 1c,d), representing a slightly harder layer assigned for the simulations (Horn and Fleige, 2003).

Fig. 2 shows the impact of the passage of the front and rear tyres of the maize harvester on the specific volume (v) and the final impact on soil bulk density after wheeling, calculated using soilDeformation(). Soil specific volume was reduced by both tyre passages (i.e. front and rear tyre effects; Fig. 2a,b), with simulated total increase in the bulk density from 1.50 to 1.57 Mg m^{-3} at a depth of 0.05 m. These results agree well with the measurements from de Lima et al. (2020b) (see confidence intervals), which found variations from 1.50 to 1.56 Mg m^{-3} at a same depth (Fig. 2c). Up to a depth of 1 m (Fig. 2c), it is possible to visualize the extension of the simulations. According to the simulations, compaction would be observed up to 0.8 m depth (shaded increase) for this simulated scenario. Furthermore, at a depth of 0.3–0.4 m, it is possible to visualize a slight deviation in the pattern of bulk density changes due to the ξ effect on the stress deviation.

The assessment of the risk of compaction using the precompression

Table 3
Assigned values for the inputs required by the functions stressTraffic() and soilDeformation()of the models from Keller (2005); Frohlich (1934), and O'Sullivan and Robertson (1996) used in the illustration examples.

Function in the soilphysics	Inputs in the *soilphysics*	Unit	Assigned value	
stressTraffic()	inflation.pressure	kPa	190 ^{front}	140 ^{rear}
	recommended.pressure	kPa	180 ^{front}	183 ^{rear}
	tyre.diameter	m	1.64 ^{front}	1.16 ^{rear}
	tyre.width	m	0.72 ^{front}	0.32 ^{rear}
	wheel.load	kg	4410 ^{front}	1100 ^{rear}
	conc.factor	–	5	5
	layers	m	0.1 to 1.0 (0.1 step)	
	stress	kPa	Calculated using the *stressTraffic()* for the assigned layers	
soilDeformation()	N	–	2.1177 ^{front}	2.0506 ^{rear,*}
	CI	–	0.0918 ^{front}	0.0849 ^{rear,*}
	k	–	0.0153 ^{front}	0.0136 ^{rear,*}
	k2	–	0.0375 ^{front}	0.0339 ^{rear,*}
	m	–	1.3 ^{front}	1.3 ^{rear}
	p.density	Mg m^{-3}	2.65 ^{front}	2.65 ^{rear}
	iBD	Mg m^{-3}	1.50 ^{front}	1.53 ^{rear}

^{front}: Front axle; ^{rear}: Rear axle. For the compressive parameters, ^{front} and ^{rear} designates the passage of the front and rear tyres. * Parameter calculated using as input the final bulk density calculated after the passage of the front tyre.

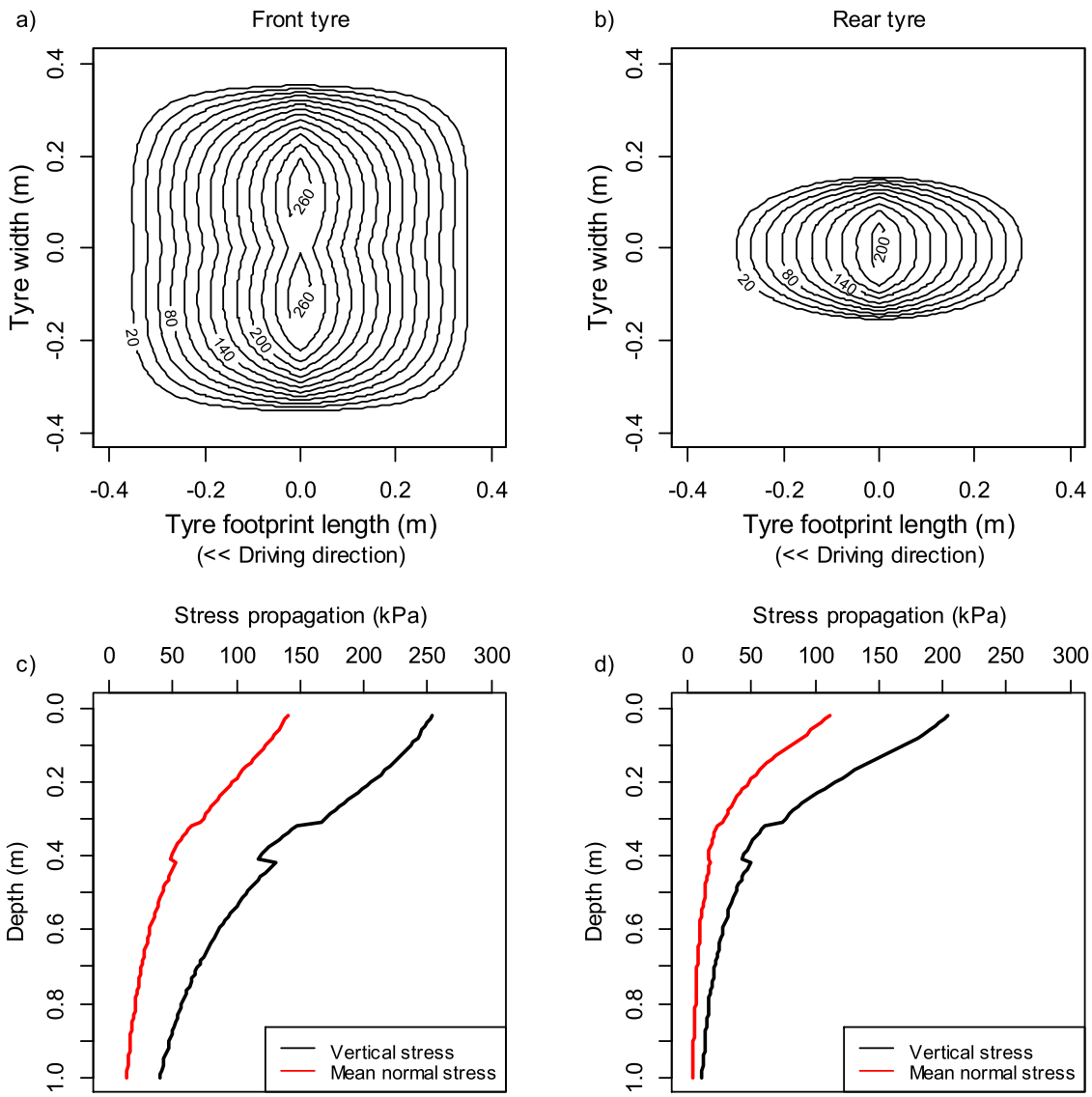

Fig. 1. Contact stress and stress propagation induced by the front (a,c) and rear (b,d) tyres of a maize harvest calculated using the function stressTraffic() with the parameters from Table 3. The stress deviation at a depth of 0.3-0.40 m exemplifies the effect of the change in the concentration factor (from 5 to 4).

stress (σ_p) is given in Fig. 3. Fig. 3a shows the mean normal stress propagation for the front tyre using σ_p as threshold for the elastic (green range) and plastic (red range) soil behaviour, whereas Fig. 3b and c show the range of risk introduced by Horn and Fleige (2003) and Stettler et al. (2014), respectively. Based on the simulations of bulk density variation presented in Fig. 2c, compaction for this scenario would be observed up to a depth of 0.8 m (shaded increase). However, Fig. 3a shows that using only σ_p as threshold criteria, compaction should be expected up to 0.30 m; below this layer, only elastic deformation should occur. Using the criterion suggested by Horn and Fleige (2003), additional plastic deformation could be expected up to 0.35 m depth (starting from the dark yellow range), for which Horn and Fleige (2009) classify the applied stress mostly too high (labile). Finally, using the criteria introduced by Stettler et al. (2014), compaction could occur up to 0.55 m, at which there would be a considerable risk of compaction (starting from the yellow range).

6. Discussion

6.1. Contact stress and stress propagation

There are several methods for estimating the stress distribution over the contact area, which are the results of the evolution of the models (Keller et al., 2007) since the pioneering approach given by Söhne (1953). Gupta and Larson (1982); Johnson and Burt (1990); O'Sullivan et al. (1999) and van den Akker (2004) described the contact area as rectangular, circular or elliptical with various forms of stress distribution. In this model/algorithm, we opted for a recent model developed by Keller (2005) because of its non-uniform stress distribution and contact area that varies in shape according to tyre parameters, besides it be calculated from readily available tyre parameters (Fig. S4). The sensitivity of the predictions of this model is discussed by Keller (2005). Comparisons with good agreement between measurements and simulations using the model from Keller (2005) can be seen in Keller et al. (2007) and Keller et al. (2016). Data from a wheeling experiment from de Lima et al. (2020a) were used to exemplify the applications of the models available in *soilphysics*, in which good agreement between measurements and simulations were observed for prediction of soil bulk

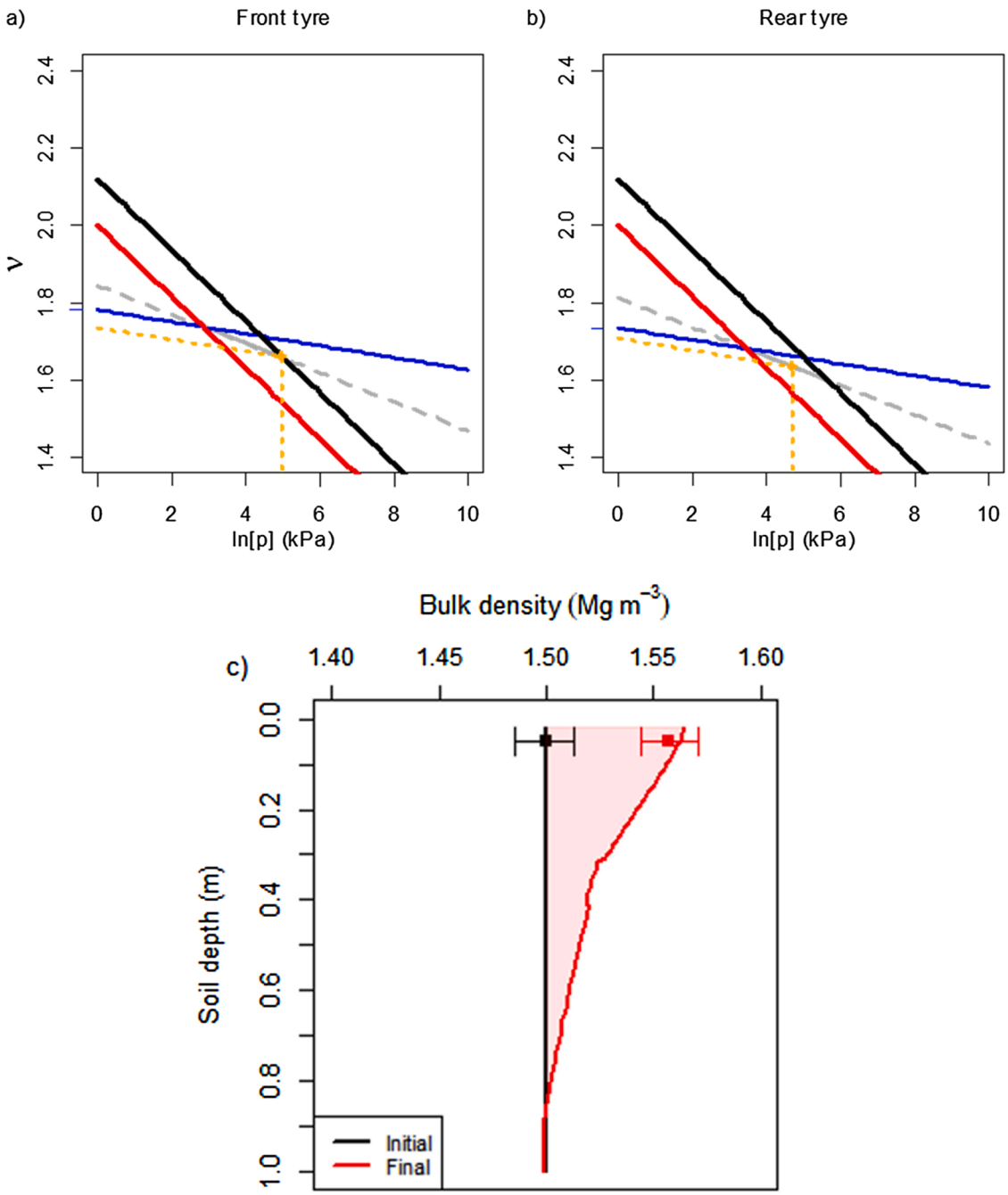

Fig. 2. Model of compression and rebound according to O'Sullivan and Robertson (1996) (soilDeformation()) exemplifying the effect on the decrease in the specific volume (*v*) after the passage of the front (a) and rear (b) tyres. The orange dashed line describes the decrease in the soil volume (in relation to blue line) as a function of the applied mean normal stress, *p*; see Fig. S3. c) Measured (at a depth of 0.05 m; see de Lima et al., 2020a) and simulated changes in soil bulk in response to applied mean normal stress, *p*, after the passage of both rear and rear tyre of a maize harvest. To illustrate stressTraffic() and soilDeformation(), changes in the soil bulk density (shaded increase) were extended to 1 m depth. Central points and bars represent the mean values and 95 % confidence intervals.

density after wheeling.

Many studies have advanced with respect to stress propagation models. Some of them have used finite element or discrete element methods to model the stress strain-relationship (e.g. Keller et al., 2014, 2016; Silva et al., 2018; Morillo and Bandeira, 2019; Pue et al., 2020). However, good agreement between field measurements and simulations with analytical models has been reported (e.g. O'Sullivan et al., 1999; Keller et al., 2007; de Lima et al., 2020a), which makes analytical models relevant in soil compaction simulations, both for accuracy and simplicity in relation to numerical models (Défossez and Richard, 2002; Keller and Lamandé, 2010). Furthermore, Keller et al. (2014, 2016) observed a similarity in the vertical stress propagation using analytical and finite element models.

The study by Keller et al. (2014) focused on an important point for the accuracy of simulations with analytical models: the concentration factor (ξ). The importance of ξ was first reported and introduced by Frohlich (1934) on the Boussinesq's (1885) equation, and was reported by Horn and Fleige (2003) as an important factor for distinguishing the mechanical behaviour of layered soils. So far, many compaction models have reported the ξ as numerical (e.g. 3, 4, 5 or 6), but dependent on a qualitative (e.g. very hard, hard, firm or soft) evaluation of soil stiffness (e.g. O'Sullivan et al., 1999; Keller et al., 2007). Keller et al. (2014)

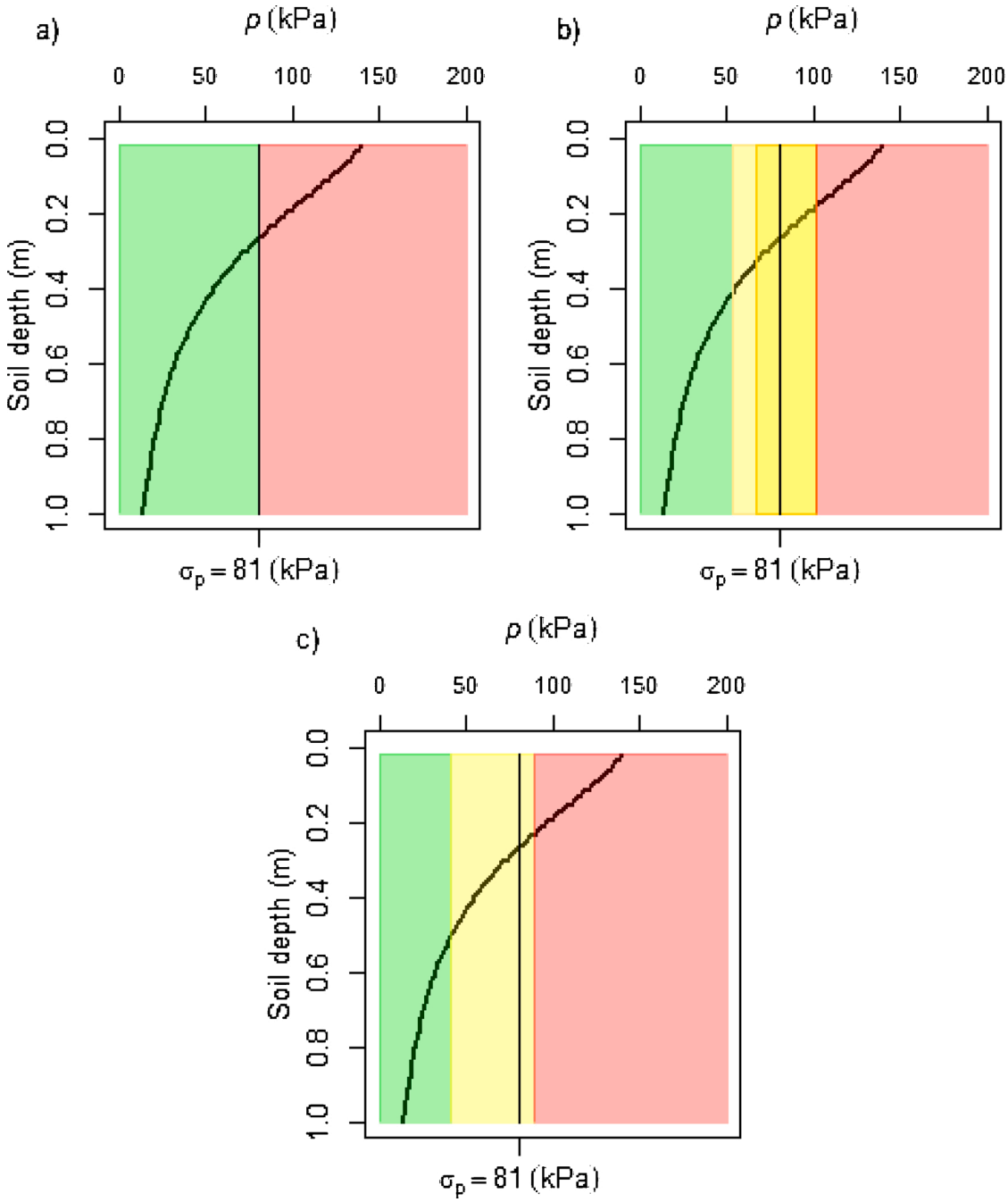

Fig. 3. Decision chart for assessment of the risk of compaction using the precompression stress (σ_p) as threshold, and the range of risk based on σ_p proposed by Horn and Fleige (2003) and Stettler et al. (2014). a) σ_p as single threshold (green range: elastic deformation, compaction is not expected; red range: plastic deformation, compaction is expected). b) Horn and Fleige (2003): $\sigma/\sigma_p > 1.5$: very stable soil, elastic deformation (green range); $\sigma/\sigma_p = 1.5$-1.2: stable soil (yellow range); $\sigma/\sigma_p =$ 1.2-0.8: labil soil (dark yellow range); and $\sigma/\sigma_p < 0.8$: weak soil, additional plastic deformation, i.e. compaction (red range). c) Stettler et al. (2014): $\sigma < 0.5\sigma_p$ (i.e. 50 % σ_p): there is no risk of compaction (green range); $\sigma > 0.5\sigma_p$ and $< 1.1\sigma_p$: considerable risk of compaction (yellow range); $\sigma > 1.1\sigma_p$ (110 % σ_p): high risk of compaction (red range).

compared finite element and analytical simulations and verified that the transmission of vertical stresses in a layered soil (i.e. with different soil stiffness) is not appreciably different from that seen in a homogeneous soil, unless very high differences in soil stiffness are considered. In other words, they verify that stress transmission was minimally affected by differences in the stiffness of a typical arable soil. The ξ was only slightly different from 3, for which the elasticity theory-based classical solution of Boussinesq (1885) is satisfied.

Horn and Fleige (2003) reported that the ξ can vary between 2 and 9, and presented a method, based on Newmark (1942), for the calculation of the ξ as function of the normal contact pressure, the radius of the tyre contact area and the soil depth. However, Horn and Fleige (2003) argue that the equation only holds true for an equivalent circular contact area. Thus, the equation introduced by Horn and Fleige (2003) seems to be an interesting way to eliminate the subjectivity of ξ, but it would need to be adapted for an oval contact area with no-uniform stress distribution. The higher the ξ, the more unstable the soil (i.e. lower σ_p) (Horn and Fleige, 2003). In our simulations, we exemplify the effect of the concentration factor for a specific soil layer by applying a lower ξ (i.e. $\xi = 4$, instead $\xi =$ 5) at a depth of 0.30−0.40 m (Fig. 1). The results were consistent with

those described by Horn and Baumgartl (2002), who argue that in stronger soils with lower values of concentration factor, the stress is transmitted more horizontally (see Horn and Fleige, 2003), reducing the stress beneath perpendicular center line. On the other hand, in weaker soils with higher concentration factor values, applied stresses are transmitted into depth, but remain concentrated around the perpendicular center line (Horn and Baumgartl, 2002). The reduction in the stress propagation beneath perpendicular center line was observed by Keller et al. (2014) for a very stiff soil layer using finite element model.

6.2. Soil volume variation and compressive parameters

According to O'Sullivan et al. (1999), besides vertical stress, compaction is also influenced by confining stresses. This means that the the state of stress is is given by the mean normal stress, p. Based on the critical state soil mechanics and using compressive parameters similar to the Cam-Clay model (Roscoe et al., 1958), O'Sullivan and Robertson (1996) formulated a model to predict the changes in the soil volume as a function of p, which is a model used to propose this algorithm, as well as that used in the soil compaction models COMPSOL (O'Sullivan et al., 1999) and *SoilFlex* (Keller et al., 2007). Therefore, the compressive parameters (e.g. N, λ_n and κ) used as inputs in the model need to be extracted from the soil compression curve as the function of p, and not from vertical stress. Thus, the simulations performed to exemplify the functionality of the function soilDeformation() were all performed in terms of mean normal stress (Fig. 2).

These compressive parameters would typically be extracted from a triaxial test, but uniaxial tests are often used because they are simpler and cheaper, and therefore available in many laboratories and accessible for many researchers (de Lima and Keller, 2019). Alternatively, some strategies can be used to estimate the p through a confined uniaxial compression test. These strategies include using the approach described by Koolen and Kuipers (1983) for a cylindrical soil sample subjected to uniaxial compression: $\sigma_2 = \sigma_3 = \sigma_1 0.5$ (where σ_1, σ_2 and σ_3 are the major, intermediate and minor principal stress). However, de Lima and Keller (2019) observed that these values should be altered due to the impact of sample dimensions, soil-cylinder wall friction and elastic properties of soil on the stress field. Alternatively, Silva et al. (2018) describe that the compressive parameters calculated in terms of vertical stress could be converted to mean normal stress by dividing their respective values by 2.3 (for more, see Silva et al., 2018). Pedo-transfer function options for estimating these compressive parameters are given in O'Sullivan et al. (1999); Défossez and Richard (2002); Keller and Arvidsson (2007) and de Lima et al. (2018) (Table 2).

6.3. Assessment of the risk of compaction

Most realistic soil compaction models include the use of p as stress acting when the soil is loaded by an agricultural tyre (O'Sullivan et al., 1999; Keller et al., 2007; de Lima et al., 2018). Thus, in addition to simulations of bulk density variation (Fig. 2), we carry out the assessment of the risk of compaction in terms of mean normal stress. The results showed that compaction should not occur at depths greater than 0.35 m using the criteria from Horn and Fleige (2003), and at depths greater than 0.55 m for the classification suggested by Stettler et al. (2014). On the other hand, simulations of bulk density variation showed that compaction could be expected up to 0.80 m, even if it is a small bulk density variation. It is interesting to note that the simulations of bulk density variation were performed using the model from O'Sullivan and Robertson (1996) (Fig. 2a and b), and in this model there is also criteria of compaction based on the value of the parameter m (Fig. S3).

Precompression stress for the simulations of bulk density variation (Fig. 2) was based on the intersection of the RCL with the VCL, and it was equal to 81 kPa. However, with the model of O'Sullivan and Robertson (1996), plastic deformations start at the yield line (YL) (Fig. S3), which in our simulations starts at 22 kPa (Fig. 2a). Thus, small increases of bulk density occurred after 22 kPa. For parameter m to be consistent with $\sigma < 0.5\sigma_p$ (there is no risk of compaction), introduced by Stettler et al. (2014), m should be assigned as $m = 0.69$, so that the plastic deformation would start at $0.5\sigma_p = 81 \times 0.5 = 40.5$ kPa. In summary, this is an important issue to improve both the risk of compaction criterion and to assign the estimate of the parameter m.

The risk of compaction assessed using σ_p has usually been addressed in terms of vertical stress, possibly because this measure of soil strength is widely studied in terms of vertical stress (e.g. Mosaddeghi et al., 2003; Imhoff et al., 2004; Saffih-Hdadi et al., 2009; Severiano et al., 2013; de Lima et al., 2015; Imhoff et al., 2016; Schjønning and Lamandé, 2018; Mendonça et al., 2020). In the algorithm proposed here, the ready calculation of the vertical stress and an available set of pedo-transfer functions (Table 2), makes the assessment of the risk of compaction in terms of vertical stress for distinct soil and sites.

6.4. Model limitations, perspectives and research needs

The model proposed here is derived from basic principles in soil physics and soil mechanics that govern compaction (e.g. stress-strain and stress-strength relationships). However, due to some more important aspects, such as shear stress and failure, shear strength components and soil displacement for prediction of ruts caused by wheel tracking, already reported in previous models to have a significant effect on tyre-soil interaction (e.g. Richards et al., 1997; Horn and Fleige, 2003; Keller et al., 2007; Stettler et al., 2014), it is possible that in some situations the model does not agree closely with field measurements. Future model updates will certainly consider the aspects listed above, besides further pedo-transfer functions and contact stress calculation options (e.g. Schjønning et al., 2008).

Research needs for better accuracy of the model should include the refinement of the range of risk of compaction which, despite some relevant efforts (e.g. Horn and Fleige, 2003, 2009; Keller et al., 2012; Stettler et al., 2014), must be further investigated. Researches with those thresholds might be extended to soil compression curve in terms of mean normal stress. Furthermore, in our illustration example, we compute the wheeling effect of both front and rear tyres on soil volume variation by correcting soil bulk density after the passage of the first tyre, but matric suction was considered as constant. The correction of matric suction for modeling the effects of repeated wheel loads may be interesting for the accuracy of the model. Some findings in this regard were carried out by Larson and Gupta (1980); Cui et al. (2010); Peth et al. (2010) and Zhai and Horn (2019).

Finally, the model limitations discussed here do not impede a simulation of soil compaction based on scientific criteria using readily available soil properties and machinery parameters. Evidently, future updates to improve the accuracy of the model with the details listed above may increase the model's intricacy and require noreadily-available inputs, which can be a disadvantage for non-expert users.

7. Computational implementation

The procedure described in Section 2 for calculation of contact stress, stress propagation, bulk density variation and the risk of compaction, was computationally implemented as a set of R functions (R software) described in Section 3, along with an interactive web application, named PredComp, both available in the R package *soilphysics* (available from https://arsilva87.github.io/soilphysics/ or http://cran.r-project.org/web/packages/soilphysics/index.html). The web app PredComp is a user-friendly version of the set of function described in Section 3.

8. Final considerations

We presented the functionalities of the package soilphysics for simulating compaction induced by agricultural field traffic using semi-analytical models. The functions were implemented giving priority to

the simplification of the steps for simulation of soil compaction, so that soil compaction models can be widely applied through an open source computational environment using readily-available soil properties and machinery parameters. The algorithms are able to provide numerical and graphical details of the tyre-soil interface (i.e. contact area), as well as the stress propagation for layered soils. Bulk density changes or risk of compaction can be calculated as responses to the applied stress. Furthermore, the soil inputs of the model can be estimated via a set of pedo-transfer functions.

Declaration of Competing Interest

The authors report no declarations of interest.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.still.2020.104824.

References

Bishwal, R.M., 2017. Potential use of R-statistical programming in the field of geoscience. 2nd International Conference for Convergence in Technology (I2CT. https://doi.org/10.1109/I2CT.2017.8226275.
Boussinesq, J., 1885. Application des potentiels á l'étude de l'équilibre et du mouvement des solides élastiques. Gautiher-Villars, Paris, 30 pp.
Cui, K., Défossez, P., Cui, Y.J., Richard, G., 2010. Soil compaction by wheeling: changes in soil suction caused by compression. Eur. J. Soil Sci. 61, 599–608.
de Lima, R.P., Keller, T., 2019. Impact of sample dimensions, soil-cylinder wall friction and elastic properties of soil on stress field and bulk density in uniaxial compression tests. Soil Till Res. 189, 15–24.
de Lima, R.P., Rolim, M.M., da Silva, A.R., Oliveira, V.S., Pedrosa, E.M.R.P., Tavares, U. E., 2015. Simple models for predicting agricultural trafficability on cohesive soils cultivated with sugarcane in Brazil. Sugar Tech 18, 347–353.
de Lima, R.P., da Silva, A.P., Giarola, N.F.B., da Silva, A.R., Rolim, M.M., Keller, T., 2018. Impact of initial bulk density and matric suction on compressive properties of two Oxisols under no-till. SoilTill Res. 175, 168–177.
de Lima, R.P., Keller, T., Giarola, N.B., Tormena, C.A., da Silva, A.R., Rolim, M.M., 2020a. Measurements and simulations of compaction effects on the least limiting water range of a no-till Oxisol. Arid. Soil Res. Rehabil. 58, 62–72.
de Lima, R.P., Rolim, M.M., Dantas, da C., da Silva, D., A. R, Mendonça, E.A., 2020b. Compressive properties and least limiting water range of plough layer and plough pan in sugarcane fields. Soil Use and Manage. https://doi.org/10.1111/sum.12601.
Défossez, P., Richard, G., 2002. Models of soil compaction due to traffic and their evaluation. Soil Till Res. 67, 41–64.
Defossez, P., Richard, G., Boizard, H., O'Sullivan, M.F., 2003. Modeling change in soil compaction due to agricultural traffic as function of soil water content. Geoderma 116, 89–105.
Frohlich, O.K., 1934. Druckverteilung im Baugrunde. Springer Verlag, Wien, 178 pp.
Gupta, S.C., Larson, W.E., 1982. Predicting soil mechanical behavior during tillage. Predicting Tillage Effects on Soil Physical Properties and Processes. American Society of Agronomy, pp. 151–178 (Special Publication 44, Chapter 10).
Horn, R., Baumgartl, T., 2002. Dynamic properties of soils. In: Warrick, A.W. (Ed.), Soil Physics Companion. CRC Press, Boca Raton, pp. 17–48.
Horn, R., Fleige, H., 2003. A method for assessing the impact of load on mechanical stability and on physical properties of soils. SoilTill Res. 73, 89–99.
Horn, R., Fleige, H., 2009. Risk assessment of subsoil compaction for arable soils in Northwest Germany at farm scale. SoilTill Res. 102, 201–208.
Horn, R., Holthusen, D., Dörner, J., Mordhorst, A., Fleige, H., 2019. Scale-dependent soil strengthening processes–what do we need to know and where to head for a sustainable environment? SoilTill Res. 195, 104388.
Imhoff, S., Silva, A.P., Fallow, D., 2004. Susceptibility to compaction, load support capacity, and soil compressibility of Hapludox. Soil Sci. Soc. Am. J. 68, 17–24.
Imhoff, S., Pires da Silva, A., Ghiberto, P.J., Tormena, C.A., Pilatti, M.A., Libardi, P.L., 2016. Physical quality indicators and mechanical behavior of agricultural soils of Argentina. PLoS One 11 (4), e0153827.
Johnson, C.E., Burt, E.C., 1990. A method of predicting soil stress state under tires. Trans. ASAE 33, 713–717.
Keller, T., 2004. Soil Compaction and Soil Tillage - Studies in Agricultural Soil Mechanics, Doctoral Thesis. Agraria 489, Swedish University of Agricultural Sciences, Uppsala, Sweden.
Keller, T., 2005. A model to predict the contact area and the distribution of vertical stress below agricultural tyres from readily available tyre parameters. Biosyst. Eng. 92, 85–96.
Keller, T., Arvidsson, J., 2007. Compressive properties of some Swedish and Danish structured agricultural soils measured in uniaxial compression tests. Eur. J. Soil Sci. 58, 1373–1381.
Keller, T., Lamandé, M., 2010. Challenges in the development of analytical soil compaction models. SoilTill Res. 111, 54–64.
Keller, T., Defossez, P., Weisskopf, P., Arvidsson, J., Richard, G., 2007. SoilFlex: a model for prediction of soil stresses and soil compaction due to agricultural field traffic including a synthesis of analytical approaches. SoilTill Res. 93, 391–411.
Keller, T., Lamandé, M., Schjønning, P., Dexter, A.R., 2011. Analysis of soil compression curves from uniaxial confined compression tests. Geoderma 163, 13–23.
Keller, T., Arvidsson, J., Schjønning, P., Lamandé, M., Stettler, M., Weisskopf, P., 2012. In situ subsoil stress-strain behavior in relation to soil precompression stress. Soil Sci. 177, 490–497.
Keller, T., Berli, M., Ruiz, S., Lamande, M., Arvidsson, J., Schjønning, P., Selvadurai, A.P. S., 2014. Transmission of vertical soil stress under agricultural tyres: comparing measurements with simulations. SoilTill Res. 140, 106–117.
Keller, T., Silva, A.P., Tormena, C.A., Giarola, N.F.B., Cavalieri, K.M.V., Stettler, M., Arvidsson, J., 2015. *SoilFlex-LLWR*: linking a soil compaction model with the least limiting water range concept. Soil Use Manage. 31, 321–329.
Keller, T., Ruiz, S., Stettler, M., Berli, M., 2016. Determining soil stress beneath a tire: measurements and simulations. Soil Sci. Soc. Am. J. 80, 541–553.
Keller, T., Sandin, M., Colombi, T., Horn, R., Or, D., 2019. Historical increase in agricultural machinery weights enhanced soil stress levels and adversely affected soil functioning. SoilTill Res. 194, 104293.
Koolen, A.J., Kuipers, H., 1983. Agricultural Soil Mechanics. Advanced Series in Agricultural Sciences, 13. Springer-Verlag, Berlin, 241 pp.
Larson, W.E., Gupta, S.C., 1980. Estimating critical stress in unsaturated soils from changes in pore water pressure during confined compression 1. Eur. J. Soil Sci. 44, 1127–1132.
Lebert, M., Horn, R., 1991. A method to predict the mechanical strength of agricultural soils. Soil Till Res. 19, 275–286.
Mendonça, E.A., de Lima, R.P., Dantas, D.D.C., Batista, P.H., Giarola, N.F., Rolim, M.M., 2020. Precompression stress in response to water content and bulk density under no-till Oxisols in southern Brazil. Geoderma Reg. 21, e00261.
Morillo, G.C.M., Bandeira, A.A., 2019. Discrete Element Method applied to the simulation of the stress state in granular materials. Arid. Soil Res. Rehabil. 57, 85–100.
Mosaddeghi, M.R., Hemmat, A., Hajabbasi, M.A., Alexandrou, A., 2003. Pre-compression stress and its relation with the physical and mechanical properties of a structurally unstable soil in central Iran. SoilTill Res. 70, 53–64.
Naderi-Boldaji, M., Hajian, A., Ghanbarian, D., Bahrami, M., 2018. Finite element simulation of plate sinkage, confined and semi-confined compression tests: a comparison of the response to yield stress. SoilTill Res. 179, 63–70.
O'Sullivan, M.F., Robertson, E.A.G., 1996. Critical state parameters from intact samples of two agricultural soils. Soil Tillage Res. 39, 161–173.
O'Sullivan, M.F., Henshall, J.K., Dickson, J.W., 1999. A simplified method for estimating soil compaction. Soil Tillage Res. 49, 325–335.
Peth, S., Rostek, J., Zink, A., Mordhorst, A., Horn, R., 2010. Soil testing of dynamic deformation processes of arable soils. Soil Tillage Res. 106, 317–328.
Pue, D.J., Lamandé, M., Schjønning, P., Cornelis, W.M., 2020. DEM simulation of stress transmission under agricultural traffic Part 3: evaluation with field experiment. SoilTill Res. 200, 104606.
R Core Team, 2020. R: A Language and Environment for Statistical Computing. URL. R Foundation for Statistical Computing, Vienna, Austria. https://www.R-project.org/.
Richards, B.G., Baumgartl, T., Horn, R., Gräsle, W., 1997. Modelling the effects of repeated wheel loads on soil profiles. Int. Agrophys. 11, 177–188.
Roscoe, K.H., Schofield, A.N., Wroth, C.P., 1958. On the yielding soils. Gkotechnique 8, 22–53.
Saffih-Hdadi, K., Defossez, P., Richard, G., Cui, Y.J., Tang, A.M., Chaplain, V.A., 2009. Method for predicting soil susceptibility to the compaction of surface layers as a function of water content and bulk density. SoilTill Res. 105, 96–103.
Schjønning, P., Lamandé, M., 2018. Models for prediction of soil precompression stress from readily available soil properties. Geoderma 320, 115–125.

Schjønning, P., Lamandé, M., Tøgersen, F.A., Arvidsson, J., Keller, T., 2008. Modelling effects of tyre inflation pressure on the stress distribution near the soil–tyre interface. Biosyst. Eng. 99, 119–133.
Severiano, E.C., Oliveira, G.C., Dias Junior, M.S., Curi, N., Costa, K.A.P., Carducci, C.E., 2013. Preconsolidation pressure, soil water retention characteristics, and texture of Latosols in the Brazilian Cerrado. Arid. Soil Res. Rehabil. 51, 193–202.
Silva, A.R., de Lima, R.P., 2015. *soilphysics*: an R package to determine soil preconsolidation pressure. Comput. Geosci. 84, 54–60.
Silva, A.R.D., Lima, R.P.D., 2016. Comparison of methods for determining precompression stress based on computational simulation. Revista Brasileira de Ciência do Solo 40 (1–10), e0150164.
Silva, R.P., Rolim, M.M., Gomes, I.F., Pedrosa, E.M.R., Tavares, U.E., Santos, A.N., 2018. Numerical modeling of soil compaction in a sugarcane crop using the finite element method. SoilTill Res. 181, 1–10.
Söhne, W., 1953. Druckverteilung im Boden und Bodenverformung unter Schlepperreifen. Grundlagen der Landtechnik 5, 49–63.
Sousa, D.F., Rodrigues, S., de Lima, H.V., Chagas, L.T., 2019. R software packages as a tool for evaluating soil physical and hydraulic properties. Comput. Electron. Agric. 168, 105077.
Stettler, M., Keller, T., Weisskopf, P., Lamandé, M., Lassen, P., Schjønning, P., 2014. Terranimo® – a web-based tool for evaluating soil compaction. Landtechnik 69, 132–137.
van den Akker, J.J.H., 2004. SOCOMO: a soil compaction model to calculate soil stresses and the subsoil carrying capacity. SoilTill Res. 79, 113–127.
Zhai, X., Horn, R., 2019. Influence of static and cyclic loading on mechanical and hydraulic properties of soils with different textures and matric potentials. Soil Soil Sci. Soc. Am. J. 83, 1601–1615.